

Encrypting and Decrypting Files with Public Key Infrastructure (3e)

Access Control and Identity Management, Third Edition - Lab 07

Student:

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Time on Task:

2 hours, 3 minutes

Progress:

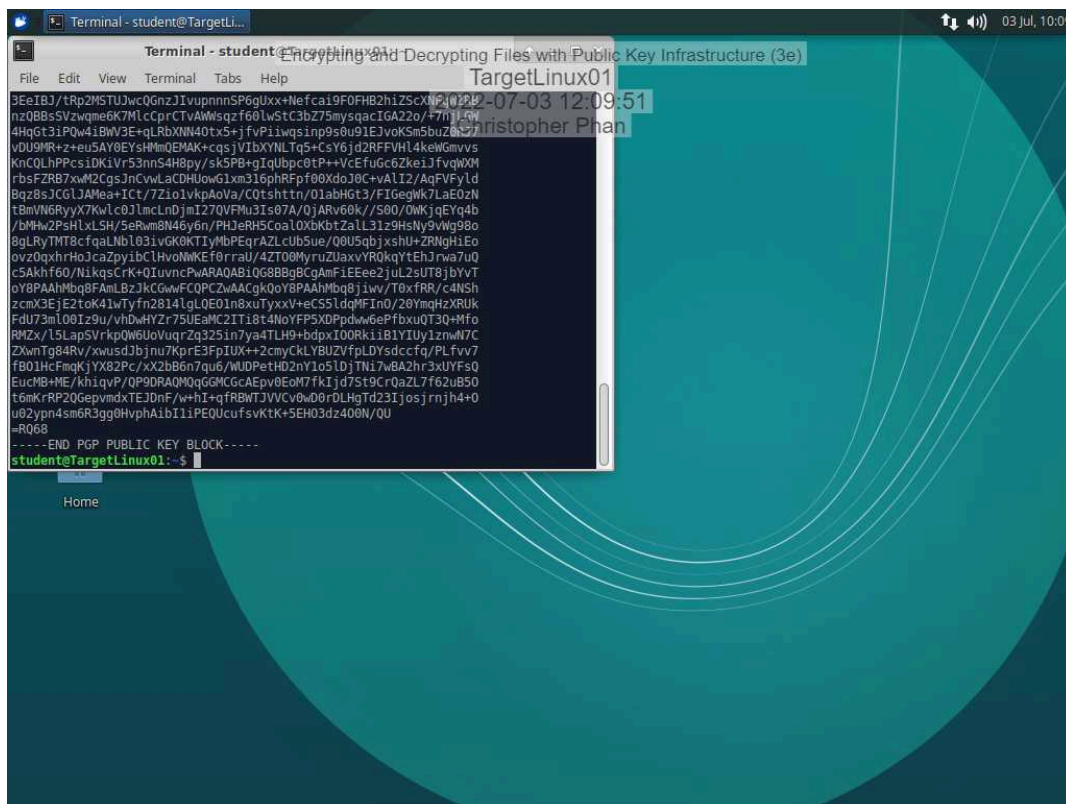
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Report Generated: Tuesday, July 5, 2022 at 9:17 PM

Section 1: Hands-On Demonstration

Part 1: Create Encryption Keys

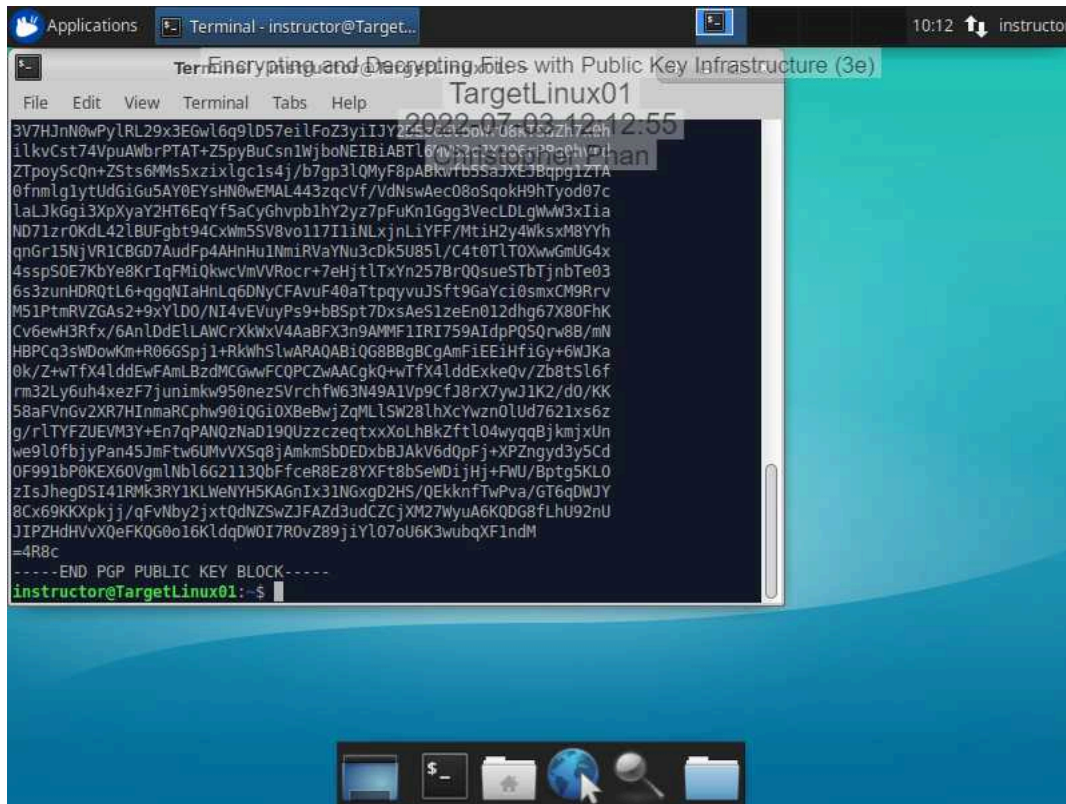
11. Make a screen capture showing Nancy's public key.



The screenshot shows a terminal window titled "Terminal - student@TargetLinux01". The terminal displays a PGP public key block for a user named "Nancy". The key is represented as a long string of ASCII characters. The terminal output is as follows:

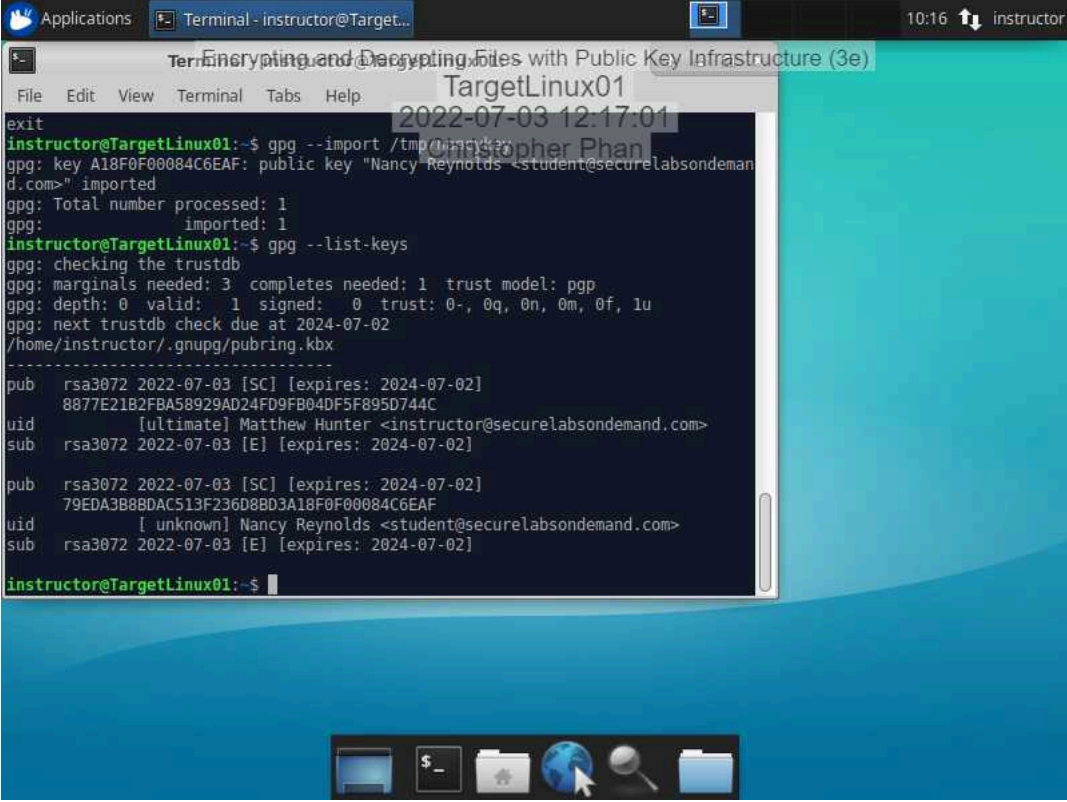
```
3EeIB3/trpMSTUJwcQgnzJIvupnnnSP6gUxx+Nefca19F0FHB2h1ZscXNFj0P2P2
nzQB8sSVzwqme6K7MlcCprCTvAWsqzF60lwStC3bZ75mysqacIGA22o/+7h1AG
4HqGt31PQw41BWV3E+qLRbXNN40tx5+jfvP11wqsinp9s0u91EJvoKSm5buZ6
vDU9NR+z+eu5AY0EYSHmQEMAK+cqsjVibXYNLtQ5+CsY6jd2RFFVHL4kewGmvvs
KnCQLhPPcs1DK1Vr53nn54H8py/sk5PB+gIqUbpcc0TP++VcEfugc6Zke1JfvqXM
rbsFZRB7xwM2CgsJncvWLaCDHUowG1xm316phRfpf00XdoJ0C+vAlI2/AqFVfyld
Bqz8sJCGLJAMea+ICt/7Zi01vKpAoVa/CQtshttn/01abHGt3/FIgegWk7LaE0zN
tBmVNRyryX7KwLc0JlmcLndJmI27QVFMu3Is07A/qjARv60k//S80/0wKjQeYq4b
/bPMhw2PsHxLSH/5eRwm8N46y6n/PHJeRHScoalOXbKbtZaLL31z9HSNy9vWg98o
BgLRyTMT8cfqaLNB1631vGK0KTIYmbPEqraZLcubSue/00U5qbJxshU+ZRNgh1Eo
ovz0qxrHrHoJcaZpyibCLHvoNikEF0rraU/4ZT00MyruZDaxvYRQqyYtEhJrwa7uQ
c5AKhf60/NI1kqsCRK+0TuvncPwARA0AB10G8BB8BcGAmFIEeezjUL2sUT8jbyVT
oYBPAAHMBqBFAmLBzjKcGwFfCQPC2wAACgk0py8PAAhMBqBjIwv/T0wfrR/c4MSH
zcmv3EjE2toK41vTyfn28141qLQEO1n8xuTyxxVveCs51dqMFI0/20YmqH2XRUK
FdlU73m100Iz9u/vhDw4Yzr75UEaMC21I18t4NoYFP5XDpDdw6ePfbxu0T30+Mfo
RMZx/15LapSVrkp0WUoVuqrZq325in7ya4TLH9+bdpxI0DRki1B1YIuy1znwI7C
ZXwmTg84Rv/xwusd3bjnu7KprE3FpIUX+2cmYkLYBUZVfpLDYsdccfq/PLfuv7
fB01HcFmqKjYX82Pc/xX2bB6n7qu6/MUDPetHD2ny1o5I0jTNI7wBA2hr3xUYFs0
EucMB+ME/khiqvP/QP9DRAQM0qG6WCGcAepv0EoM7fK1jd75t9Cr0aZL7f62uB50
t6mKrRP20GepvmdxTE3DnF/w+hi+qfRBWTJVVcV0w00DLHgTd23josjrnjh4+0
u02ypn4sm6R3gg0HvphAibI11PEQUcufsvKtk+SEH03dz400N/QU
=R068
-----END PGP PUBLIC KEY BLOCK-----
student@TargetLinux01:~$
```

21. Make a screen capture showing Matthew's public key.



Part 2: Encrypt a File

12. Make a screen capture showing the output of the `--list-keys` command.



The screenshot shows a terminal window titled "Terminal - instructor@TargetLinux01" with a blue background. The terminal displays the following commands and output:

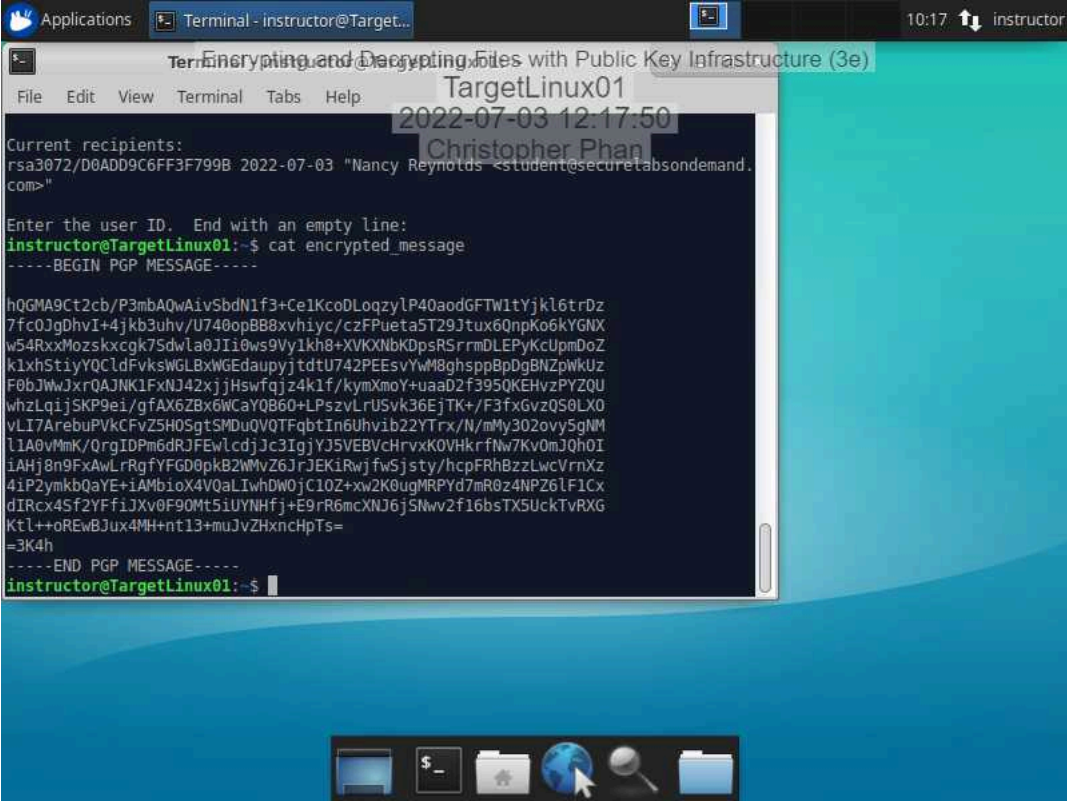
```
exit
instructor@TargetLinux01:~$ gpg --import /tmp/pubring.kbx
gpg: key A18F0F00084C6EAF: public key "Nancy Reynolds <student@securelabsondemand.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid: 1  signed: 0  trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2024-07-02
/home/instructor/.gnupg/pubring.kbx
-----
pub   rsa3072 2022-07-03 [SC] [expires: 2024-07-02]
      8877E21B2FBA58929AD24F09FB04DF5F895D744C
uid       [ultimate] Matthew Hunter <instructor@securelabsondemand.com>
sub   rsa3072 2022-07-03 [E] [expires: 2024-07-02]

pub   rsa3072 2022-07-03 [SC] [expires: 2024-07-02]
      79EDA3B8BDAC513F236D8B03A18F0F00084C6EAF
uid       [ unknown] Nancy Reynolds <student@securelabsondemand.com>
sub   rsa3072 2022-07-03 [E] [expires: 2024-07-02]

instructor@TargetLinux01:~$
```

The terminal window is overlaid on a desktop environment with a blue background. The desktop has a taskbar at the bottom with icons for a terminal, a file manager, a globe, a magnifying glass, and a folder. The terminal window title bar includes the text "Applications", "Terminal - instructor@TargetLinux01", and a system clock showing "10:16" and the username "instructor".

18. Make a screen capture showing the **encrypted message**.



The screenshot shows a terminal window titled "Terminal - instructor@TargetLinux01" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The terminal output is as follows:

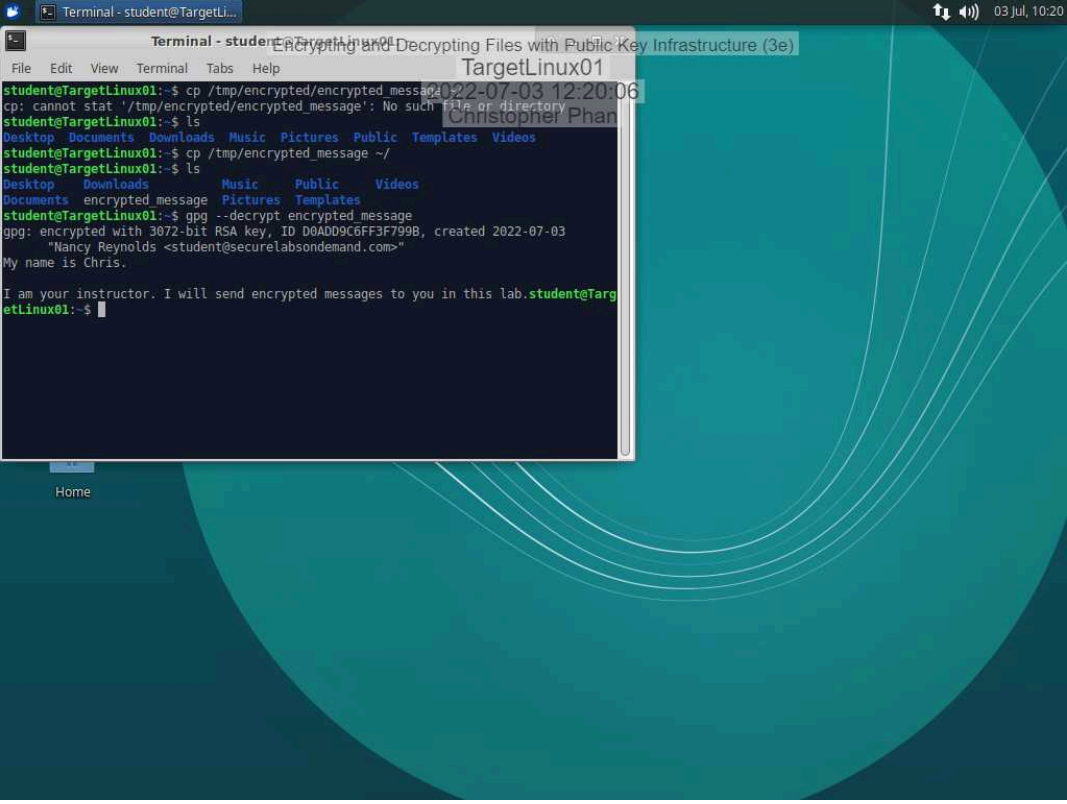
```
Current recipients:
rsa3072/D0ADD9C6FF3F799B 2022-07-03 "Nancy Reynolds <student@securetabsondemand.com>"

Enter the user ID. End with an empty line:
instructor@TargetLinux01:~$ cat encrypted_message
-----BEGIN PGP MESSAGE-----

hQGMA9Ct2cb/P3mbA0wAivSbdN1f3+Ce1KcoDLoqzylP40aodGFTW1tyjkl6trDz
7fc0JgDhvI+4jkb3uhv/U740pBB8xvhiyc/czFPueta5T29Jtux6QnpK06kYGNX
w54RxxMozskxcgk7SdwlaoJIi0ws9Vy1kh8+XVXKNbK0psRSrrmDLEPyKcUpmDoZ
k1xhStiyYQClDFvksWGLBxWGEdaupyjtdtU742PEEsvYwMBghsppBpDgBNZpwkUz
F0bJWwJxrQAjNK1FxNJ42xjHswfjz4k1f/kymXmoY+uaa02f395QKEHvzPYZOU
whzLqjSKP9ei/gfAX6ZBx6WCaYQB60+LPszvLRusvk36EjTK+/F3fxGvzQS0LX0
vLI7ArebuPVkCFvZ5H0SgtSMDuQVQTFqbtIn6Uhb22YTrx/N/mMy302ovy5gNM
l1A0vMmK/QrgIDPm6dRjFEwLcdjJc3IgYJ35VEBVcHrvxKOVHkrfNw7Kv0mJQhOI
iAHj8n9FxAwLrRgfYFGD0pkB2NMvZ6JrJEKiRwjfwSjsty/hcpFRhBzzLwcVrnXz
4iP2ymkbQaYE+iAMbioX4VQaLIwhDW0jC10Z+Xw2K0ughMRPYd7mR0z4NPZ6lF1Cx
dIRcx4Sf2YFfiJXv0F90Mt5iUYNHfj+E9rR6mcXNJ6j5Nwv2f16bsTX5UckTvRXG
KtL++oREwBJux4MH+nt13+muJvZHxncHpTs=
=3K4h
-----END PGP MESSAGE-----
instructor@TargetLinux01:~$
```

Part 3: Decrypt a File

3. Make a screen capture showing the decrypted message.



```
Terminal - student@TargetLinux01
File Edit View Terminal Tabs Help
student@TargetLinux01:~$ cp /tmp/encrypted/encrypted_message ~/
cp: cannot stat '/tmp/encrypted/encrypted_message': No such file or directory
student@TargetLinux01:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
student@TargetLinux01:~$ cp /tmp/encrypted_message ~/
student@TargetLinux01:~$ ls
Desktop  Downloads  encrypted_message  Music  Public  Videos
Documents  encrypted_message  Pictures  Templates
student@TargetLinux01:~$ gpg --decrypt encrypted_message
gpg: encrypted with 3072-bit RSA key, ID D0ADD9C6FF3F799B, created 2022-07-03
"Nancy Reynolds <student@securelabsondemand.com>"
My name is Chris.

I am your instructor. I will send encrypted messages to you in this lab.
student@TargetLinux01:~$
```

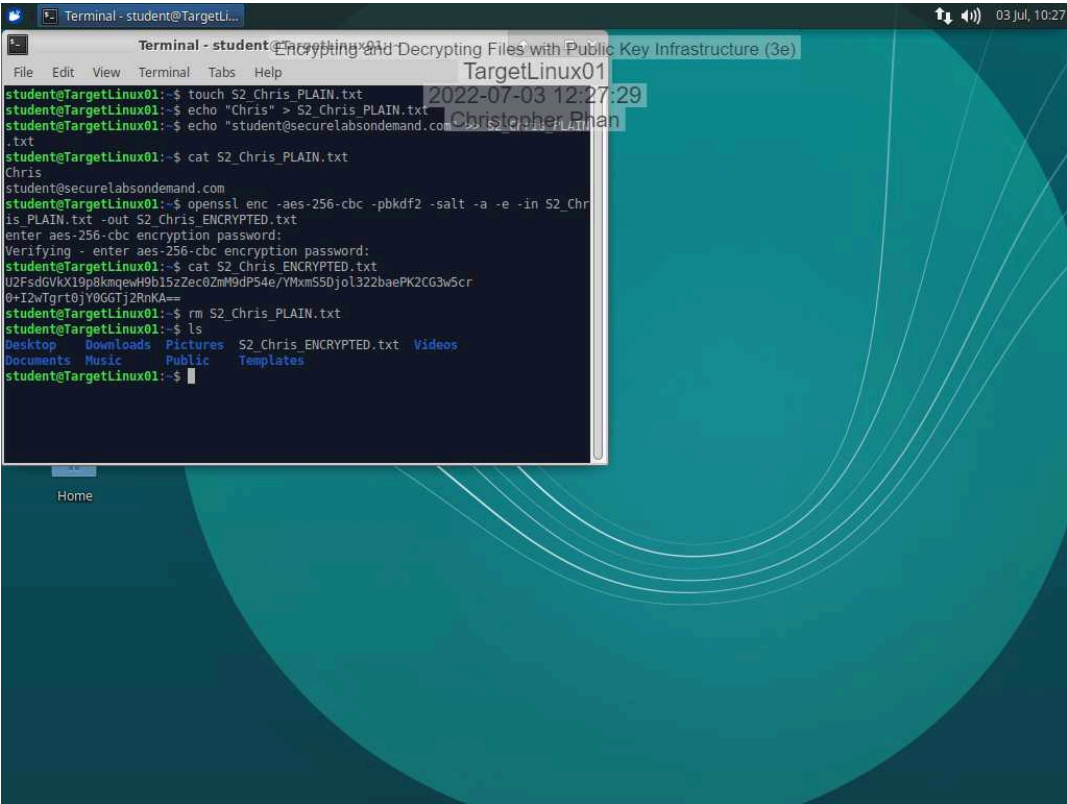
Section 2: Applied Learning

Part 1: Encrypt a File with Symmetric Encryption

10. In the Lab Report file, **document** the password you used to encrypt the file.

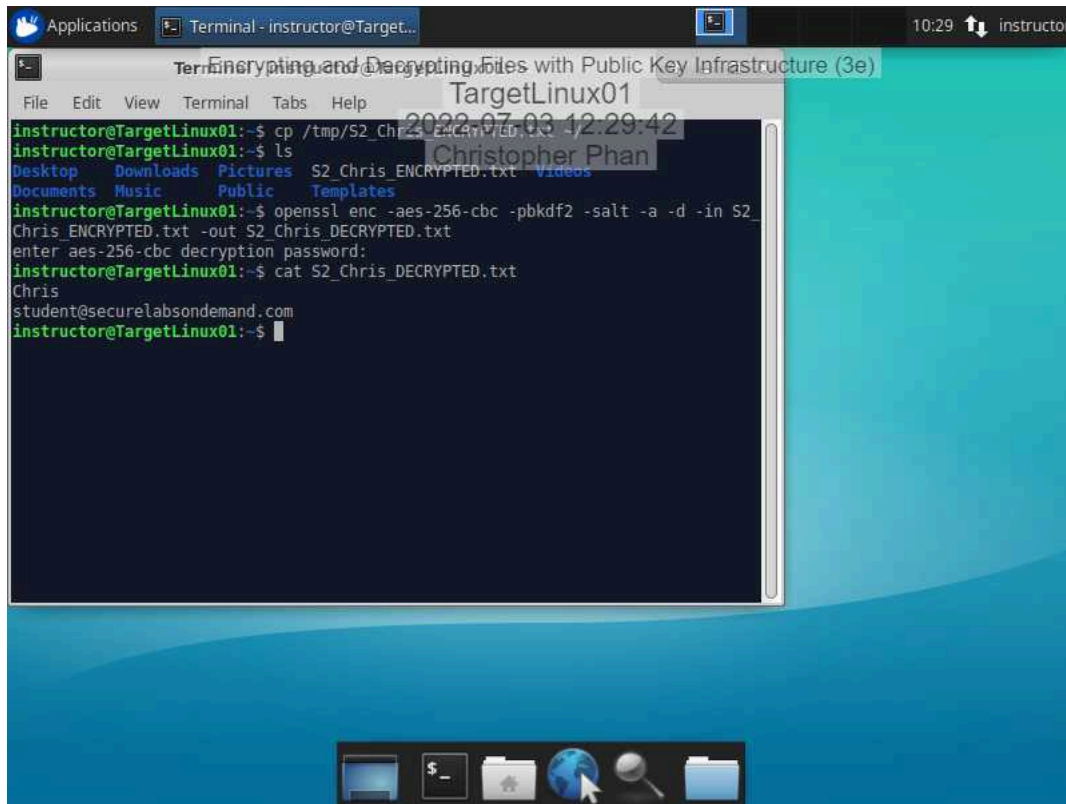
Password = "yourownpassword"

14. **Make a screen capture** showing the **output of the ls command**.



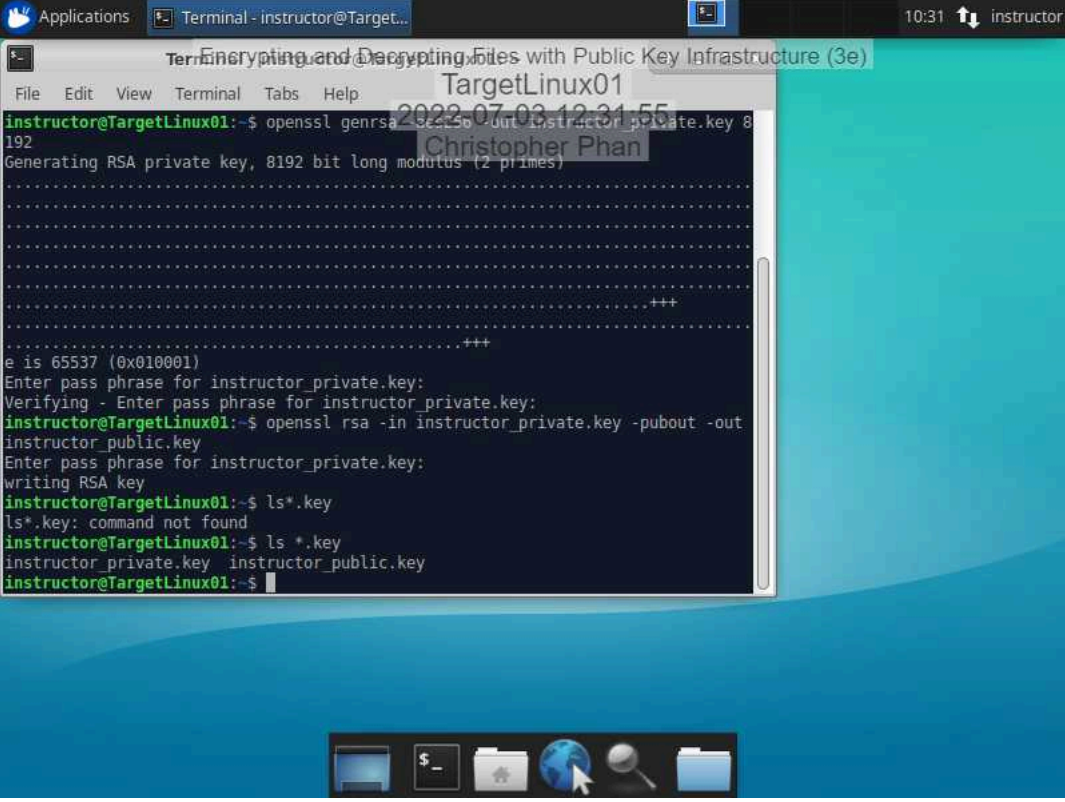
```
student@TargetLinux01:~$ touch S2_Chris_PLAIN.txt
student@TargetLinux01:~$ echo "Chris" > S2_Chris_PLAIN.txt
student@TargetLinux01:~$ cat S2_Chris_PLAIN.txt
Chris
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in S2_Chris_PLAIN.txt -out S2_Chris_ENCRYPTED.txt
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat S2_Chris_ENCRYPTED.txt
U2FsZG9kX19p8kmgewH9b15zZec0ZmH9dP54e/YMxmS5Djol322baePK2CG3wScr
0+I2wTgrt0jY0GGTj2RnKA==
student@TargetLinux01:~$ rm S2_Chris_PLAIN.txt
student@TargetLinux01:~$ ls
Desktop  Downloads  Pictures  S2_Chris_ENCRYPTED.txt  Videos
Documents  Music      Public    Templates
```


25. Make a screen capture showing the contents of the decrypted file.



Part 2: Encrypt a File with Asymmetric Encryption

7. Make a screen capture showing the instructor's key pair files.



The screenshot shows a terminal window titled "Terminal - instructor@TargetLinux01" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The terminal output shows the following commands and results:

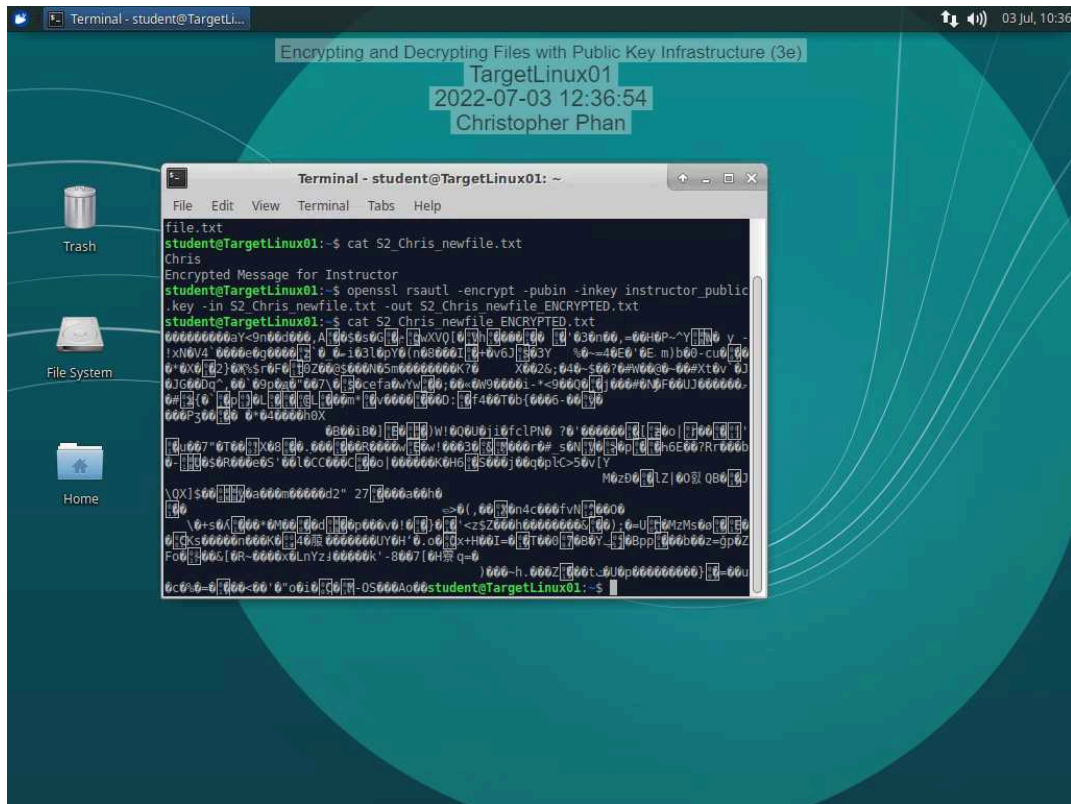
```
instructor@TargetLinux01:~$ openssl genrsa -des3 -out instructor_private.key 8192
Generating RSA private key, 8192 bit long modulus (2 primes)
.....
.....+++
.....+++
e is 65537 (0x010001)
Enter pass phrase for instructor_private.key:
Verifying - Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ openssl rsa -in instructor_private.key -pubout -out instructor_public.key
Enter pass phrase for instructor_private.key:
writing RSA key
instructor@TargetLinux01:~$ ls*.key
ls*.key: command not found
instructor@TargetLinux01:~$ ls *.key
instructor_private.key  instructor_public.key
instructor@TargetLinux01:~$
```

The terminal window is overlaid on a desktop background with a blue gradient. A taskbar at the bottom contains icons for a terminal, file manager, web browser, and search. A watermark "TargetLinux01" and a timestamp "2022-07-03 12:31:55" are visible over the terminal output.

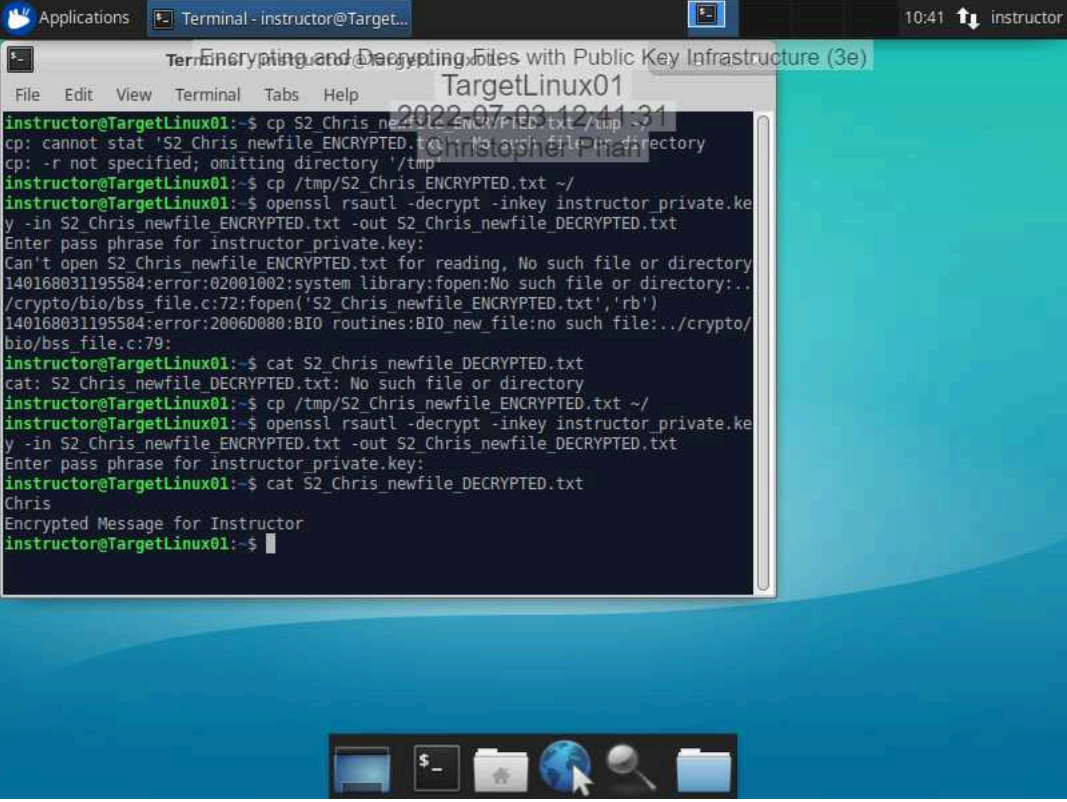
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20. **Make a screen capture** showing the encrypted contents of the **S2_yourname_newfile_ENCRYPTED.txt** file.



30. Make a screen capture showing the **decrypted contents of the S2_yourname_newfile_DECRYPTED.txt** file.



```
instructor@TargetLinux01:~$ cp S2_Chris_newfile_ENCRYPTED.txt /tmp/S2_Chris_newfile_ENCRYPTED.txt
cp: cannot stat 'S2_Chris_newfile_ENCRYPTED.txt': No such file or directory
cp: -r not specified; omitting directory '/tmp'
instructor@TargetLinux01:~$ cp /tmp/S2_Chris_ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.key -in S2_Chris_newfile_ENCRYPTED.txt -out S2_Chris_newfile_DECRYPTED.txt
Enter pass phrase for instructor_private.key:
Can't open S2_Chris_newfile_ENCRYPTED.txt for reading, No such file or directory
140168031195584:error:02001002:system library:fopen:No such file or directory:../crypto/bio/bss_file.c:72:fopen('S2_Chris_newfile_ENCRYPTED.txt','rb')
140168031195584:error:2006D080:BIIO routines:BI0_new_file:no such file:../crypto/bio/bss_file.c:79:
instructor@TargetLinux01:~$ cat S2_Chris_newfile_DECRYPTED.txt
cat: S2_Chris_newfile_DECRYPTED.txt: No such file or directory
instructor@TargetLinux01:~$ cp /tmp/S2_Chris_newfile_ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.key -in S2_Chris_newfile_ENCRYPTED.txt -out S2_Chris_newfile_DECRYPTED.txt
Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ cat S2_Chris_newfile_DECRYPTED.txt
Chris
Encrypted Message for Instructor
instructor@TargetLinux01:~$
```

Section 3: Challenge and Analysis

Make a screen capture showing the **commands used to encrypt and copy the file**.

Incomplete

Make a screen capture showing the **commands used to copy and decrypt the file**.

Incomplete

Make a screen capture showing the **decrypted file contents**.

Incomplete